

Lecture 1

Magic cloaks

Based on a lecture of Sergei Ivanov [13].

A Scattering data

Given a Riemannian manifold M denote by $\tau: SM \rightarrow M$ its unit tangent bundle

$$SM = \{v \in TM : |v| = 1\}.$$

Recall that by Liouville's theorem, the geodesic flow $\varphi^t: SM \rightarrow SM$ preserves the natural volume on SM in its domain of definition. Denote by G the vector field on SM that defines the geodesic flow φ^t .

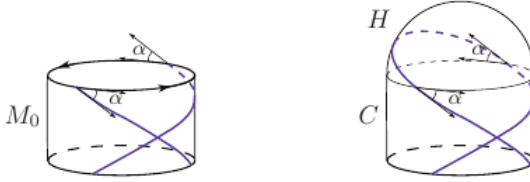
Suppose M has nonempty boundary ∂M ; in other words, M is a closed region of an ambient Riemannian manifold bounded by a smooth hypersurface ∂M . Denote by $\partial_+ SM$ the set of unit vectors at points on ∂M that point in M . Note that $\partial_+ SM$ is a bundle over ∂M with fibers formed by closed half-spheres. The set $\partial_+ SM$ is a subset of ∂SM that can be also defined as the closure of the subset at which the geodesic flow enters SM . The full boundary ∂SM is a sphere bundle over ∂M ; it contains $\partial_+ SM$ as a subbundle.

Consider a geodesic γ_u in the direction of a vector $u \in \partial_+ SM$. In other words, $\gamma_u(t) = \tau \circ \varphi^t(u)$. Suppose that γ_u hits the boundary again. Denote by $\ell(u)$ the first hitting time, so $\gamma_u(\ell(u)) \in \partial M$. Note that in this case the vector $v = -\gamma'(\ell)$ lies in $\partial_+ SM$. The map $s: u \mapsto v$ is defined if $\ell(u) < \infty$; so s is a partially defined involution on SM which we will call scattering map.

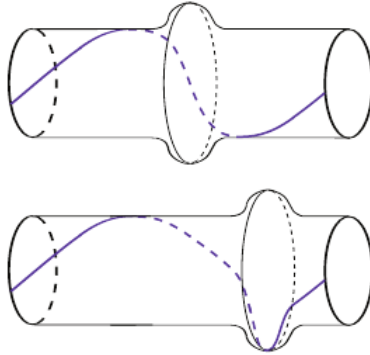
Suppose that M and $\bar{M}$ be two compact connected Riemannian manifolds with boundary such that a neighborhood of ∂M can be isometrically identified with a neighborhood of $\partial \bar{M}$, and moreover, the

scattering maps in M and $\bar{M}$ are identical. In this case we say that M and $\bar{M}$ have identical scattering data. If in addition their hitting time functions coincide (that is, if $\ell(u) \equiv \bar{\ell}(u)$ for any $u \in \partial_+ SM$), then we say that M and $\bar{M}$ have the identical lens data.

Notice that if a manifold contains a copy of a round hemisphere, then cutting it and gluing the opposite points of its boundary produces a manifold with identical scattering data. The lens data for the



constructed pair of manifolds are not identical. An example of nonisometric manifolds with identical lens data can be found among surfaces of revolution which look like cylinders with bumps on them that are shifted and otherwise look the same.



1.1. Exercise. Check that the described examples have identical lens data.

Construct a pair of nonisometric Riemannian metrics on the disc with identical lens data.

1.2. Exercise. Show that the above construction does not work if instead of bump we use a dimple.

B Main theorem

The following theorem proved by Mikhael Gromov [10]; it is the main subject of this lecture.

1.3. Theorem. *Any connected compact region M of Euclidean space of dimension at least 2 cut by a smooth hypersurface is scattering rigid; that is, any manifold with scattering data identical to M is isometric to M .*

1.4. Corollary. *Suppose a Riemannian metric g on $\mathbb{R}^n$ coincides with Euclidean metric g_0 outside of a compact set K . Suppose that the complement $\gamma \setminus K$ of any nontrivial g -geodesic γ coincides with the complement of a line (as sets). Then $(\mathbb{R}^n, g)$ is isometric to the Euclidean space.*

Note that in the corollary we cannot claim that $g = g_0$. Indeed for any diffeomorphism $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$ that is identical outside K the metric $g = \varphi^* g_0$ satisfies the assumption in the corollary. Clearly one can choose φ so that $g \neq g_0$.

C Identical lens data

1.5. Lemma. *Suppose that a manifold $\bar{M}$ has identical scattering data with a compact region M in Euclidean space of dimension at least 2. Then M and $\bar{M}$ have identical lens data.*

The Euclidean space can be exchanged to any complete Riemannian manifold with unique geodesic between any pair of points; the proof is the same.

Proof. Denote by W the complement of the interior of M in the Euclidean space E . Let us glue $\bar{M}$ to W along the isometry in the definition of scattering data. This way we obtain a complete Riemannian manifold $\bar{E}$ that is Euclidean outside of a $\bar{M}$.

Suppose that a smooth hypersurface Σ in E surrounds M . Then Σ cuts from E and $\bar{E}$ manifolds with identical scattering data. Moreover if $\bar{M}$ and M are not isometric, then the obtained pair is not isometric as well.

It follows that we can assume that M is a ball.¹ In this case any geodesic $\bar{\gamma}$ in $\bar{E}$ visits $\bar{M}$ at most once. In other words, if $\bar{\gamma}$ enters $\bar{M}$, then the complement $\bar{\gamma} \setminus \bar{M}$ has two connected components which are parts of a line γ in E .

¹This is true for the proof of the lemma and theorem as well.

Choose a unit-speed geodesic $\bar{\gamma}$ that visits $\bar{M}$. Let us include it in a smooth one-parameter family of unit-speed geodesics $\bar{\gamma}_\tau$ for $\tau \in [0, 1]$ so that $\bar{\gamma}_0$ does not visit $\bar{M}$ and $\bar{\gamma}_1 = \bar{\gamma}$. (Here we use that $\dim M \geq 2$.)

We can assume that the vector field $\bar{\mathbf{I}} = \frac{\partial}{\partial \tau} \bar{\gamma}_\tau(t)$ is orthogonal to $\bar{\gamma} = \frac{\partial}{\partial t} \bar{\gamma}_\tau(t)$ at every point $\bar{\gamma}_\tau(t_0)$ for some fixed t_0 .

Observe that in this case $\langle \bar{\mathbf{I}}, \bar{\mathbf{T}} \rangle = 0$ at all points $\bar{\gamma}_\tau(t)$. Indeed

$$\begin{aligned} \frac{\partial}{\partial t} \langle \bar{\mathbf{I}}, \bar{\mathbf{T}} \rangle &= \bar{\mathbf{T}} \langle \bar{\mathbf{I}}, \bar{\mathbf{T}} \rangle = \\ &= \langle \nabla_{\bar{\mathbf{T}}} \bar{\mathbf{I}}, \bar{\mathbf{T}} \rangle + \langle \bar{\mathbf{I}}, \nabla_{\bar{\mathbf{T}}} \bar{\mathbf{T}} \rangle = \\ &= \langle \nabla_{\bar{\mathbf{T}}} \bar{\mathbf{T}}, \bar{\mathbf{T}} \rangle = \\ &= \frac{1}{2} \cdot \bar{\mathbf{I}} \langle \bar{\mathbf{T}}, \bar{\mathbf{T}} \rangle = \\ &= \frac{\partial}{\partial \tau} |\bar{\gamma}'_\tau(t)|^2 = \\ &= 0. \end{aligned}$$

That is, $\langle \bar{\mathbf{I}}, \bar{\mathbf{T}} \rangle$ does not depend on t . Since $\langle \bar{\mathbf{I}}, \bar{\mathbf{T}} \rangle = 0$ at $\bar{\gamma}_\tau(t_0)$, the same holds for all points $\bar{\gamma}_\tau(t)$.

Consider a family of geodesics γ_τ in E that coincide with $\bar{\gamma}_\tau$ (as sets) outside of M . The same argument shows that $\langle \mathbf{I}, \mathbf{T} \rangle = 0$ for the corresponding vector fields $\mathbf{I} = \frac{\partial}{\partial \tau} \gamma_\tau(t)$ and $\mathbf{T} = \frac{\partial}{\partial t} \gamma_\tau(t)$ at all points $\gamma_\tau(t)$.

By assumption, $\bar{\mathbf{T}} = \mathbf{T}$ and $\bar{\mathbf{I}} - \mathbf{I}$ is proportional to $\mathbf{T}$ in W . It follows that $\bar{\mathbf{I}} = \mathbf{I}$ in W . Therefore $\gamma_\tau(t) = \bar{\gamma}_\tau(t)$ for any t and τ , provided that $\bar{\gamma}(t) \in W$. Whence the γ spends exactly the same time in M as $\bar{\gamma}$ spends in $\bar{M}$ and the lemma follows. $\square$

Comments. The identity $\langle \mathbf{I}, \mathbf{T} \rangle = 0$ is proved the same way as the Gauss lemma. The vector fields as $\mathbf{I}$ is an example of the so-called Jacobi fields along γ_τ ; they describe a variation of a geodesics, and we will see them again.

1.6. Exercise. Let M be a connected manifold with boundary. Assume that hitting time in M is bounded by ℓ ; that is, any geodesic that has one end on ∂M has length at most ℓ . Show that any geodesic in M has length at most ℓ .

Hint: Show that the set of vectors $\mathbf{U} \in SM$ such that the arc $\gamma_{\mathbf{U}}|_{[0, \ell]}$ lies in M is open and closed; here the geodesic $\gamma_{\mathbf{U}}$ defined by $\gamma'_{\mathbf{U}}(0) = \mathbf{U}$.

D Volume equality

1.7. Lemma. Suppose that a manifold $\bar{M}$ has identical scattering data with a compact region M in Euclidean space of dimension at

least 2. Then

$$\text{vol } M = \text{vol } \bar{M}.$$

Proof. We will denote by $\bar{\tau}: S\bar{M} \rightarrow \bar{M}$ the unit tangent bundle over $\bar{M}$ and by $\bar{\varphi}^t: S\bar{M} \rightarrow S\bar{M}$ its the geodesic flow for time t .

Set $\bar{\Omega} = \bar{\tau}^{-1}(\bar{M})$. Since geodesic flow preserves the volume, we get

$$\begin{aligned} \text{vol } \mathbb{S}^{n-1} \cdot \text{vol}(\bar{M}, g) &= \text{vol } \bar{\Omega} = \\ &= \text{vol}[\bar{\varphi}^t(\bar{\Omega})] \end{aligned}$$

for any t .

By 1.6, we can choose t so that $\tau(v) \notin \bar{M}$ for any $v \in \varphi^t(\bar{\Omega})$.

Repeat the same construction for M . By 1.5 and 1.6, $\varphi^t(\Omega) = \varphi^t(\bar{\Omega})$. In particular,

$$\text{vol } \Omega = \text{vol}[\varphi^t(\Omega)] = \text{vol}[\bar{\varphi}^t(\bar{\Omega})] = \text{vol } \bar{\Omega}.$$

whence the result follows. $\square$

E Santaló formula

Santaló formula is a corollary of Liouville's theorem — geodesic flow preserves the volume. It gives an expression for a volume of a Riemannian manifold with boundary in terms of hitting times of its geodesics. It provides a more direct proof of 1.7.

Suppose M is a Riemannian manifold with nonempty boundary ∂M . Recall that

- ◊ SM denotes the unit tangent bundle over M .
- ◊ φ^t denotes geodesic flow. In particular, if γ_u is the geodesic in M defined by $\gamma'_u(0) = u$, then $\gamma'_u(t) = \varphi^t(u)$.
- ◊ $\ell: SM \rightarrow [0, \infty]$ denotes the hitting time of γ_u in the boundary of M .
- ◊ $\partial_+ SM$ denotes by the bundle of unit vectors at points on ∂M that point in M . It can be defined as the closure of the subset of ∂SM at which the geodesic flow enters SM .

1.8. Theorem. *Let M be an n -dimensional Riemannian manifold with nonempty boundary. Suppose that any unit-speed geodesic in M hits its boundary in finite time. Then for any smooth function $f: SM \rightarrow \mathbb{R}$ the following identity holds:*

$$\int_{w \in SM} f(w) = \int_{u \in \partial_+ SM} \langle u, N \rangle \cdot \int_0^{\ell(u)} f \circ \varphi^t(u) \cdot dt,$$

where $\mathbf{N}$ denotes the unit vector field normal to ∂M that points inside M .

By taking $f \equiv 1$ in the above theorem, we get the following.

1.9. Corollary. *If M is as in the theorem, then*

$$\text{vol } \mathbb{S}^{n-1} \cdot \text{vol } M = \int_{\mathbf{u} \in \partial_+ SM} \ell(\mathbf{u}) \cdot \langle \mathbf{u}, \mathbf{N} \rangle.$$

Proof. Since any unit-speed geodesic in M hits its boundary in finite time, any vector $\mathbf{w} \in SM$ can be uniquely described as $\varphi^t(\mathbf{u})$ for some $\mathbf{u} \in \partial_+ SM$ and $0 \leq t \leq \ell(\mathbf{u})$. In other words SM can be identified with the subgraph

$$\Phi = \{ (\mathbf{u}, t) \in (\partial_+ SM) \times \mathbb{R} : 0 \leq t \leq \ell(\mathbf{u}) \}.$$

The subgraph Φ has two volume forms: the first, say ω , is the pull back of the volume form on SM ; the second $\chi = dt \wedge \alpha$, where α is the volume form on ∂SM .

Note that both forms are invariant with respect to vertical shifts (along $\mathbb{R}$). For ω it is true by Liouville's theorem; for χ it follows from the definition.

Set $r(\mathbf{v}) = \text{dist}_{\partial M} \circ \tau(\mathbf{v})$; note that r is a smooth function near ∂SM . Observe that $dr = \langle \mathbf{u}, \mathbf{N} \rangle \cdot dt$ on ∂SM . Note that the equality $\omega = dr \wedge \alpha$ holds on ∂SM . Whence

$$\textbf{1} \quad \omega = \langle \mathbf{u}, \mathbf{N} \rangle \cdot \chi$$

on ∂SM . Since both forms ω and α are invariant with respect to vertical shifts, we get that **1** holds everywhere in Φ . $\square$

1.10. Exercise. *Construct two Riemannian metrics g_0 and g_1 on the disc $\mathbb{D}$ that coincide near the boundary and such that*

$$\text{area}(\mathbb{D}, g_0) > \text{area}(\mathbb{D}, g_1),$$

but

$$\ell_0(\mathbf{u}) < \ell_1(\mathbf{u}),$$

where $\ell_i(\mathbf{u})$ denotes hitting time of g_i -geodesic in the direction $\mathbf{u}$; that is, $\ell_i(\mathbf{u})$ is the length of g_i -geodesic that starts at a point $p \in \partial \mathbb{D}$ in the direction $\mathbf{u} \in \partial_+ S\mathbb{D}$.

Why does this example not contradict the Santaló formula?

1.11. Exercise. Denote by ω the volume form on SM and by G the vector field on SM that describes the geodesic flow. Given a function $f: SM \rightarrow \mathbb{R}$, consider the function $F: SM \rightarrow \mathbb{R}$ defined by

$$F(u) = - \int_0^{\ell(u)} f \circ \varphi^t(u) \cdot dt.$$

Prove the Santaló formula applying Stokes' theorem to form $\iota_G(F \cdot \omega)$.

F Differentiability of distance function

1.12. Theorem. For any closed set A in a complete Riemannian manifold M and any point $x \notin A$ the differential $d_x f$ of the distance function $f = \text{dist}_A$ is defined if and only if there is a unique minimizing geodesic γ from x to A .

Moreover, if a unit vector $u \in T_x$ points in the direction of the unique geodesic γ , then

$$d_x f(w) = -\langle u, w \rangle$$

for any $w \in T_x$; or, equivalently,

$$\nabla_x f = -u.$$

Proof; only-if part. Choose a closed set A , a point $x \notin A$. Let γ be a unit-speed minimizing geodesic from x to A ; denote by $y \in A$ the other end of γ .

Choose a smooth unit-speed curve α that such that $\alpha(0) = x$, and set $w = \alpha'(0)$. Observe that

$$\begin{aligned} |\gamma(\tfrac{t}{\varepsilon}) - \alpha(t)|_M &= t \cdot \sqrt{\tfrac{1}{\varepsilon^2} - 2 \cdot \langle u, w \rangle \cdot \tfrac{1}{\varepsilon} + 1} + o(t) = \\ &= \tfrac{1}{\varepsilon} \cdot t - \langle u, w \rangle \cdot t + O(\varepsilon \cdot t) \end{aligned}$$

for any fixed $\varepsilon > 0$. Applying the triangle inequality for y , $\gamma(\tfrac{t}{\varepsilon})$, and $\alpha(t)$, we get that

$$\begin{aligned} f \circ \alpha(t) &= |y - \alpha(t)|_M \leq \\ &\leq |y - \gamma(\tfrac{t}{\varepsilon})|_M + |\gamma(\tfrac{t}{\varepsilon}) - \alpha(t)|_M \leq \\ &\leq |p - x| - t \cdot \langle u, w \rangle + O(\varepsilon \cdot t). \end{aligned}$$

Passing to the limit as $\varepsilon \rightarrow 0$, we get

$$\textcircled{1} \quad (f \circ \alpha)'(0) \leq -\langle u, w \rangle$$

if the left hand side is defined.

Observe that if $d_x f$ is defined, then $(f \circ \alpha)'(0) = d_x f(w)$. Therefore

$$d_x f(w) \leq -\langle u, w \rangle$$

for any $w \in T_x$. Since both sides of the last inequality are linear in w , we get that the equality

$$d_x f(w) = -\langle u, w \rangle$$

holds for any $w \in T_x$.

Suppose that γ_1 is another minimizing geodesic from x to A ; set $u_1 = \gamma_1'(0)$. If $d_x f$ is defined, then we have

$$-\langle u, w \rangle = d_x f(w) = -\langle u_1, w \rangle;$$

that is, $u_1 = u$ and therefore $\gamma_1 = \gamma$.

If part. Suppose that γ is a unique geodesic from x to A . Choose α as above. For each t choose a minimizing geodesic γ_t from $\alpha(t)$ to A ; Set $u(t) = \gamma_t'(0)$ and $w(t) = \alpha'(t)$.

Recall that f and α are Lipschitz. By Rademacher's theorem and $\textcircled{1}$, we have that

$$(f \circ \alpha)'(t) \stackrel{\text{a.e.}}{=} -\langle u(t), w(t) \rangle;$$

moreover

$$f \circ \alpha(\tau) - f \circ \alpha(0) = - \int_0^\tau \langle u(t), w(t) \rangle \cdot t.$$

It remains to show that $\langle u(t), w(t) \rangle \rightarrow \langle u, w \rangle$ as $t \rightarrow 0$.

The latter follows if $u(t) \rightarrow u$ as $t \rightarrow 0$. Assume the contrary, then there is a sequence $t_n \rightarrow 0$ such that $u(t_n)$ converges to a unit vector $v \in T_x$ that is distinct from u . The minimizing geodesics γ_{t_n} converge to a geodesic from x to A that runs in the direction v . Since $v \neq u$, this geodesic is distinct from γ — a contradiction. $\square$

1.13. Exercise. Suppose that M is a compact Riemannian manifold with convex boundary ∂M ; that is, any shortest path in M may only have its endpoints on ∂M . Assume that for any $p \in \partial M$ the function dist_p is differentiable on $\partial M \setminus \{p\}$.

Prove the following statements:

- (a) Any geodesic in M is minimizing.
- (b) For any $p \in M$ the distance function dist_p is differentiable in $M \setminus \{p\}$.
- (c) Show that M is homeomorphic to a ball.
- (d) The restriction of the distance function to ∂M determines the lens data of M .

G Besicovitch inequality

The following theorem was proved by Abram Besicovitch [1].

1.14. Theorem. *Let g be a metric tensor on a unit n -dimensional cube $\square$. Suppose that the g -distances between the opposite facets of $\square$ are at least 1; that is, any Lipschitz curve that connects opposite faces has g -length at least 1. Then $\text{vol}(\square, g) \geq 1$.*

The following statement is assumed to be known.

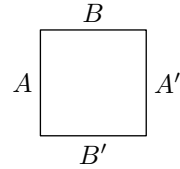
1.15. Area inequality. *Let $f: \mathcal{M} \rightarrow \mathcal{N}$ be a locally Lipschitz map between n -dimensional Riemannian manifolds. Suppose that $|\text{jac}_p f| \leq 1$ almost everywhere in A , then*

$$\text{vol } A \geq \text{vol}[f(A)].$$

Proof. We will consider the case $n = 2$; the other cases are proved the same way.

Denote by A , A' , and B , B' the opposite sides of the square $\square$. Consider two functions

$$\begin{aligned} f_A(x) &:= \min\{\text{dist}_A(x)_g, 1\}, \\ f_B(x) &:= \min\{\text{dist}_B(x)_g, 1\}. \end{aligned}$$



Let $f: \square \rightarrow \square$ be the map with coordinate functions f_A and f_B ; that is, $f(x) := (f_A(x), f_B(x))$.

Observe that f maps each face to itself. Indeed,

$$x \in A \implies \text{dist}_A(x)_g = 0 \implies f_A(x) = 0 \implies f(x) \in A.$$

Similarly, if $x \in B$, then $f(x) \in B$. Further,

$$x \in A' \implies \text{dist}_A(x)_g \geq 1 \implies f_A(x) = 1 \implies f(x) \in A'.$$

Similarly, if $x \in B'$, then $f(x) \in B'$.

Therefore

$$f_t(x) = t \cdot x + (1 - t) \cdot f(x)$$

defines a homotopy of maps of the pair of spaces $(\square, \partial\square)$ from f to the identity map. It follows that degree of f is 1; that is, f sends the fundamental class of $(\square, \partial\square)$ to itself. In particular f is onto.

Suppose that Jacobian matrix $\text{Jac}_p f$ of f is defined at $p \in \square$. Choose an orthonormal frame in T_p with respect to g and the standard frame in the target $\square$. Observe that the differentials $d_p f_A$ and $d_p f_B$ written in these frames are the rows of $\text{Jac}_p f$. Evidently $|d_p f_A| \leq 1$ and $|d_p f_B| \leq 1$. Since the determinant of a matrix is the volume of the parallelepiped spanned on its rows, we get

$$|\text{jac}_p f| \leq |d_p f_A| \cdot |d_p f_B| \leq 1.$$

Since $f: \square \rightarrow \square$ is a Lipschitz onto map, the area inequality (1.15) implies that

$$\text{vol}(\square, g) \geq \text{vol} \square = 1.$$

□

The following theorem can be proved along the same lines.

1.16. Theorem. *Let (M, g) be an n -dimensional Riemannian manifold. Suppose that there is a degree 1 map from its boundary ∂M to the surface of n -dimensional cube $\square$; denote by $d_1, \dots, d_n$ the distances between the inverse images of pairs of opposite facets of $\square$ in ∂M . Then*

$$\text{vol}(M, g) \geq d_1 \cdots d_n.$$

1.17. Exercise. *Suppose g is a metric tensor on a regular hexagon $\diamond$ such that g -distances between the opposite sides are at least 1. Is there a positive lower bound on $\text{area}(\diamond, g)$?*

1.18. Exercise. *Let V be a compact set in the n -dimensional Euclidean space $\mathbb{E}^n$ bounded by a hypersurface Σ . Suppose g is a Riemannian metric on V such that*

$$|p - q|_g \geq |p - q|_{\mathbb{E}^n}$$

for any two points $p, q \in \Sigma$. Show that

$$\text{vol}(V, g) \geq \text{vol}(V)_{\mathbb{E}^n}.$$

H Equality case

1.19. Theorem. *Suppose that equality holds in 1.16, then $\text{vol}(M, g)$ is isometric to the product $[0, d_1] \times \cdots \times [0, d_n]$.*

Proof. We will prove the 2-dimensional case, assuming that $d_1 = d_2 = 1$; the general case can be proved along the same lines. Let us use the same notation as in the proof of 1.14.

Consider the map $s: x \mapsto (\text{dist}_A(x)_g, \text{dist}_B(x)_g)$. From the proof of 1.14 we get that $\text{Im } s \supset \square$. Observe that in the case of equality we have that $\text{Im } s = \square$. Indeed, the same argument shows that

$$\text{vol}(s^{-1}(\square), g) \geq \text{vol } \square = 1.$$

The set $s^{-1}(\mathbb{R}^2 \setminus \square)$ is an open subset of $\square$. If it is nonempty, then it has positive volume. In this case

$$\text{vol}(\square, g) > \text{vol}(s^{-1}(\square), g) \geq 1$$

— a contradiction.

Summarizing: there is a geodesic path of unit g -length connecting any point on one side of the square to a point on the opposite side.

Moreover, for any pair of opposite facets and an interior point $p \in \square$, there is a unique geodesic path of unit g -length from one face to the other that passes thru p . The latter can be shown by cutting $\square$ into two rectangles by a level set of dist_A thru p , applying the above statement to both rectangles and taking the concatenation of the obtained geodesic paths with end at p . If such a path is not unique, then one could make a shortcut near p — a contradiction.

Let γ be such a geodesic path from A to A' . By 1.12, $\gamma'(t) = \nabla_{\gamma(t)} \text{dist}_A$. Therefore dist_A is differentiable at every point $p \in \square$. It follows that the map s is differentiable.

Further, checking the equality case in each inequality in the proof of 1.14, we get that s is a bijection and the equalities

$$|d_p \text{dist}_A| = 1, \quad |d_p \text{dist}_B| = 1, \quad \text{and} \quad \langle d_p \text{dist}_A, d_p \text{dist}_B \rangle = 0$$

hold for almost all $p \in \square$. Since $d_p \text{dist}_A$ and $d_p \text{dist}_B$ are well defined, we get that the equalities hold everywhere. That is, s is an isometry. $\square$

I Proof assembling

Proof of 1.3. Suppose that $\bar{M}$ and M have identical scattering data. By 1.5 $\bar{M}$ and M have identical lens data. Further, by 1.7 (or by

Santaló formula 1.8), we have

$$\text{vol } M = \text{vol } \bar{M}.$$

Without loss of generality we may assume that M lies in a unit cube $\square$. Cut from $\square$ the manifold M and glue in $\bar{M}$ by the isometry provided by the definition of scattering data; denote the obtained modified cube by $\bar{\square}$. Note that the distances between points on the boundary of $\bar{\square}$ remain unchanged. The latter follows that distance is the length of a minimizing geodesic between a pair of points and the geodesics in $\square$ and $\bar{\square}$ behave exactly the same way and they spend exactly the same time in M and $\bar{M}$ respectively.

It follows that in the Besicovitch inequality, an equality holds for $\bar{\square}$. By 1.19, $\bar{\square}$ is isometric to $\square$; whence $\bar{M}$ is isometric to M . $\square$

J More exercises

Two Riemannian metrics g_0 and g_1 on M are called conformally equivalent if there is a function λ such that $g_1 = \lambda^2 \cdot g_0$. In this case the function λ is called conformal factor. Note that for any g_0 -unit-speed curve $\gamma: [a, b] \rightarrow M$ we have

$$\text{length}_{g_1} \gamma = \int_a^b \lambda \circ \gamma(t) \cdot dt$$

1.20. Exercise. Let g_0 be the canonical metric on the projective space $\mathbb{RP}^n$; that is, $(\mathbb{RP}^n, g_0)$ is isometric to the quotient space of the unit sphere $\mathbb{S}^n$ by central symmetry. Suppose that g_1 is conformally equivalent to g_0 . Denote by ℓ_0 and ℓ_1 the systoles — the lengths of shortest noncontractible closed curves in $(\mathbb{RP}^n, g_0)$ and $(\mathbb{RP}^n, g_1)$ respectively (so $\ell_0 = \pi$). Show that

$$\frac{\text{vol}(\mathbb{RP}^n, g_1)}{\ell_1^n} \geq \frac{\text{vol}(\mathbb{RP}^n, g_0)}{\ell_0^n}.$$

Hint: Use that geodesic flow preserves volume of the unit tangent bundle to rewrite the integral of conformal factor over $(\mathbb{RP}^n, g_0)$ and interpret the result.

1.21. Definition. A compact Riemannian manifold M with nonempty boundary ∂M is called *simple* if any geodesic in M is minimizing and its boundary is convex; that is, any shortest path in M may only have its endpoints on ∂M .

Note that 1.13 provides a condition on a manifold with boundary that guarantees its simplicity.

1.22. Exercise. Let (M, g_0) be a simple Riemannian manifold. Suppose that a conformally equivalent metric $g_1 = \lambda^2 \cdot g_0$ on M induce the same distances on the boundary ∂M ; that is,

$$|x - y|_{g_1} = |x - y|_{g_0}$$

for any $x, y \in \partial M$. Show that $\lambda \equiv 1$; that is, $g_1 = g_0$.

Hint: Apply 1.13 plus the Santaló formula 1.8 and argue similarly to 1.20.

1.23. Conjecture. Let (M, g_0) be a simple Riemannian manifold and g_1 is another Riemannian metric on M . Suppose that the metric induced by g_1 on ∂M is at least as large as the metric induced by g_0 ; that is,

$$|x - y|_{g_1} \geq |x - y|_{g_0}$$

for any $x, y \in \partial M$. Then

$$\text{vol}(M, g_1) \geq \text{vol}(M, g_0).$$

Let (M, g) be a Riemannian manifold. The Sasaki metric is a natural choice of Riemannian metric $\hat{g}$ on the total space of the tangent bundle $\tau: TM \rightarrow M$ defined the following way:

Identify the tangent space $T_u[TM]$ for any $u \in T_p M$ with the direct sum of vertical and horizontal subspaces $T_p M \oplus T_p M$. The projection of this splitting is defined by the differential $d\tau: TTM \rightarrow TM$ and we assume that the velocity of a curve in TM formed by a parallel field along a curve in M is horizontal. Then $T_u[TM]$ is equipped with the metric $\hat{g}$ defined by

$$\hat{g}(X, Y) = g(X^V, Y^V) + g(X^H, Y^H),$$

where X^V and $X^H \in T_p M$ denote the vertical and horizontal components of $X \in T_u[TM]$.

1.24. Exercise. Let g be the canonical Riemannian metric on the sphere S^2 . Consider the tangent bundle TS^2 equipped with the induced Sasaki metric $\hat{g}$. Let S_R be the hypersurface in TS^2 of vectors with norm R ; we assume that S_R is equipped with induced Riemannian metric.

Show that $\text{vol } S_R \rightarrow \infty$ as $R \rightarrow \infty$, but $\text{diam } S_R$ stays bounded for all R .

K Remarks

The fact that not all manifolds are scattering rigid was pointed out by Christopher Croke [5]; jointly with Bruce Kleiner, he constructed more examples [6].

Theorem 1.3 has a number of variations and generalizations. In particular an analog of this theorem holds in the following cases:

- ◊ For regions in 2-dimensional Riemannian manifolds with unique geodesic between any two points; proved by Leonid Pestov and Gunther Uhlmann [18].
- ◊ For regions in a round hemispheres; proved by René Michel [15].
- ◊ For regions in hyperbolic spaces; it follows from the result of Gérard Besson, Gilles Courtois, and Sylvestre Gallot [2].
- ◊ For regions in the product space $\mathbb{R} \times M$, where M is a Riemannian manifold with unique geodesic between any two points; proved by Christopher Croke and Bruce Kleiner [7].
- ◊ For small regions in any Riemannian manifold; proved by Dmitri Burago and Sergei Ivanov [3].

The fundamental theorem of Riemannian geometry often refers to the existence and uniqueness of Levi-Civita connection. But it also refers to the result of John Nash [17] that *any Riemannian manifold is isometric to a smooth submanifold in a Euclidean space*. In particular, Riemannian manifold can be defined as a submanifold of Euclidean space with induced metric tensor.

We will prove Nash's theorem for compact manifolds (1.28). This case is leading; the general case requires a sequence of additional elementary statements that is outlined in Exercise ???.

In the proof we use the so-called Günther trick — an essential simplification of the original proof that was found by Matthias Günther [11]. We also use several minor improvements found by Deane Yang, Terence Tao, and Ralph Howard [12, 20, 21].

L Induced metric

Choose a smooth n -dimensional manifold Ω ; here and further, smooth means C^∞ .

Recall that a field of bilinear symmetric forms, say g , on the tangent bundle $T\Omega$ is called metric tensor. A metric tensor g is called Riemannian if it is positive; that is, if $g(x, x) > 0$ for any tangent vector $x \neq 0$.

We will consider $\mathbb{R}^q$ with standard Euclidean metric. Recall that derivative of $w: \Omega \rightarrow \mathbb{R}^q$ in direction of a tangent vector $x \in T_p \Omega$ is denoted by

$$xw = dw(x).$$

Let $w: \Omega \rightarrow \mathbb{R}^q$ be a smooth map. We say that g is induced metric tensor for w if

$$g(x, y) = \langle xw, yw \rangle$$

for any two tangent vectors $x, y \in T_p \Omega$ at any point $p \in \Omega$.

More generally, suppose $v, w: \Omega \rightarrow \mathbb{R}^q$ are two smooth maps. Let us write $g = Q(v, w)$ if

$$g(x, y) \equiv \frac{1}{2} \cdot [\langle xv, yw \rangle + \langle xw, yv \rangle].$$

The operator Q is symmetric and bilinear; it takes two smooth maps $v, w: \Omega \rightarrow \mathbb{R}^q$ and spits a metric tensor $g = Q(v, w)$ on Ω .

Note that if v and w are C^k -smooth, then $Q(v, w)$ is C^{k-1} -smooth. Observe that *a metric tensor g is induced by a smooth map $w: \Omega \rightarrow \mathbb{R}^q$ if and only if $g = Q(w, w)$* .

The components of $g = Q(w, w)$ in a local chart can be written as

$$g_{ij} = Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle.$$

1.25. Exercise. *Let w be a smooth map from a smooth Riemannian manifold (Ω, g) to the Euclidean space $\mathbb{R}^q$. Show that g is induced by w if and only if w is length-preserving; that is, if*

$$\text{length } \gamma = \text{length}(w \circ \gamma)$$

for any curve γ in Ω .

The map $w: (\Omega, g) \rightarrow \mathbb{R}^q$ as in the exercise are often called isometric, but this term might be for other types of maps.

Let Ω be a smooth manifold. Given two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{q_2}$, denote by $w = v_1 \oplus v_2$ the map

$$w: \Omega \rightarrow \mathbb{R}^{q_1+q_2} = \mathbb{R}^{q_1} \oplus \mathbb{R}^{q_2}$$

defined by $w: p \mapsto (v_1(p), v_2(p))$.

1.26. Exercise. *Show that*

$$Q(v_1 \oplus v_2, v_1 \oplus v_2) = Q(v_1, v_1) + Q(v_2, v_2)$$

for any two smooth maps $v_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $v_2: \Omega \rightarrow \mathbb{R}^{q_2}$.

Recall that support of a map is the closure of the set of its arguments with nonzero values.

1.27. Exercise. *Let $v_1, v_2: \Omega \rightarrow \mathbb{R}^q$ be smooth maps defined on a smooth manifold Ω . Suppose v_1 and v_2 have disjoint support. Show that*

$$Q(v_1 + v_2, v_1 + v_2) = Q(v_1 \oplus v_2, v_1 \oplus v_2).$$

M Formulation

The following theorem is the main result of this lecture.

1.28. Theorem. *Let (Ω, g) be a compact Riemannian manifold. Then, for sufficiently large integer q , there is a smooth immersion $w: \Omega \rightarrow \mathbb{R}^q$ such that*

$$g = Q(w, w).$$

1.29. Exercise. *Construct a metric tensor g on the torus $\mathbb{T}^2 = \mathbb{S}^1 \times \mathbb{S}^1$ such that $g(X, X) \geq 0$ for any tangent vector X , but there is no smooth map $w: \mathbb{T}^2 \rightarrow \mathbb{R}^{10}$ with the induced metric g .*

1.30. Exercise. Apply 1.28 and the Whitney embedding theorem to show that in 1.28 one can assume that w is an embedding. In other words, any compact Riemannian manifold is isometric to a submanifold in a Euclidean space.

N Free maps

As before, Ω is a smooth n -dimensional manifold.

Recall that a C^1 map $f: \Omega \rightarrow \mathbb{R}^q$ is called regular if its differential df has rank n at each point. In other words, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all n first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ are linearly independent.

A C^2 map $f: \Omega \rightarrow \mathbb{R}^q$ is called free (also known as two-regular) if an analogous property holds for first *and second* partial derivatives; that is, for any point $p \in \Omega$ and some (and therefore any) local coordinates at p all $n + \frac{n \cdot (n+1)}{2}$ partial derivatives $\partial_i f, \partial_{ij} f$ for $i \leq j$ are linearly independent.

1.31. Exercise. Show that there is no free map $\mathbb{S}^2 \rightarrow \mathbb{R}^4$.

1.32. Exercise. Let $f_1: \Omega \rightarrow \mathbb{R}^{q_1}$ and $f_2: \Omega \rightarrow \mathbb{R}^{q_2}$ be smooth maps defined on a smooth manifold Ω . Suppose f_1 is free; show that so is $f_1 \oplus f_2$.

Let f be a smooth function on a manifold Ω . Denote by E_f the smooth map $\Omega \rightarrow \mathbb{R}^2$, defined by

$$E_f(x) = (\cos[f(x)], \sin[f(x)]).$$

In other words, E_f is the composition of two maps

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell} \mathbb{S}^1 \subset \mathbb{R}^2,$$

where $\ell: \mathbb{R} \rightarrow \mathbb{S}^1$ is the natural covering maps $y \mapsto (\cos y, \sin y)$.

1.33. Exercise. Consider the (x, y) -plane $\mathbb{R}^2$. Show that the map

$$E_x \oplus E_y \oplus E_{x+y}: \mathbb{R}^2 \rightarrow \mathbb{R}^6 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2$$

is free.

Generalize the statement to maps $\mathbb{R}^n \rightarrow \mathbb{R}^{n \cdot (n+1)}$.

1.34. Exercise. Let $f: \Omega \rightarrow \mathbb{R}^s$ be a regular smooth map and $F: \mathbb{R}^s \rightarrow \mathbb{R}^q$ is free. Show that the composition $F \circ f: \Omega \rightarrow \mathbb{R}^q$ is free.

Apply the Whitney embedding theorem and 1.33 to conclude that any smooth manifold admits a free embedding into a Euclidean space.

O Key lemmas: formulations

In this section we recall the definition of Hölder norm for functions on torus and formulate the Nash's and Günther's lemma.

Let us denote by $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ the n -dimensional torus. The torus comes with a global system of periodic coordinates which can be used to define the partial derivatives $\partial_1, \dots, \partial_n$. A metric tensor on $\mathbb{T}^n$ can be also written in these coordinates; so we can (and will) think that metric tensor is smooth map $g: \mathbb{T}^n \rightarrow \mathbb{R}^N$; here and further

$$N := \frac{n \cdot (n+1)}{2}.$$

Choose $0 < \alpha < 1$, say $\alpha = \frac{1}{2}$. Denote by $|x - y|_{\mathbb{T}^n}$ the standard distance between points $x, y \in \mathbb{T}^n$. Let us recall the definition of Hölder norms of a function $u: \mathbb{T}^n \rightarrow \mathbb{R}$

$$\begin{aligned} \|u\|_{0,\alpha} &:= \sup_{x \in \mathbb{T}^n} \{ |u(x)| \} + \sup_{x, y \in \mathbb{T}^n} \left\{ \frac{|u(x) - u(y)|}{|x - y|_{\mathbb{T}^n}^\alpha} \right\}, \\ \|u\|_{k,\alpha} &:= \max_{|\mathcal{I}| \leq k} \{ \|\partial_{\mathcal{I}} u\|_{0,\alpha} \}, \end{aligned}$$

where $\mathcal{I} = (i_1, \dots, i_n)$ denotes multi-index; so, $\partial_{\mathcal{I}} := \partial_1^{i_1} \dots \partial_n^{i_n}$ and $|\mathcal{I}| := i_1 + \dots + i_n$. The space of functions with finite norm $\| \cdot \|_{k,\alpha}$ is called the (k, α) -Hölder space; it is denoted by $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

1.35. Exercise. Let $v \in C^\infty(\mathbb{T}^n, \mathbb{R})$. Show that there is a constant c such that

$$\|u \cdot v\|_{k,\alpha} \leq c \cdot \|u\|_{k,\alpha}$$

for any $u \in C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$.

Further, let us define $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ as the space of functions $u: \mathbb{T}^n \rightarrow \mathbb{R}^q$ with all coordinate functions from $C^{k,\alpha}(\mathbb{T}^n, \mathbb{R})$. Let

$$\|u\|_{k,\alpha} := \max_i \{ \|u_i\|_{k,\alpha}, \dots, \|u_q\|_{k,\alpha} \},$$

where $u_1, \dots, u_q$ are the coordinate functions of u .

In the following two lemmas we fix a smooth free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ and let

$$L(w) := 2 \cdot Q(f, w).$$

1.36. Nash's lemma. The linear operator

$$L: C^\infty(\mathbb{T}^n, \mathbb{R}^q) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^N)$$

has a right inverse

$$M: C^\infty(\mathbb{T}^n, \mathbb{R}^N) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^q)$$

such that

$$\|Mh\|_{2,\alpha} \leq a \cdot \|h\|_{2,\alpha}$$

for some constant a .

1.37. Günther's lemma. *There is a symmetric bilinear operator*

$$\tilde{Q}: C^\infty(\mathbb{T}^n, \mathbb{R}^q) \times C^\infty(\mathbb{T}^n, \mathbb{R}^q) \rightarrow C^\infty(\mathbb{T}^n, \mathbb{R}^q)$$

such that

$$L\tilde{Q} = Q,$$

$$\|\tilde{Q}(w, w)\|_{2,\alpha} \leq b_2 \cdot \|w\|_{2,\alpha}^2,$$

$$\|\tilde{Q}(w, w)\|_{k,\alpha} \leq b_2 \cdot \|w\|_{2,\alpha} \cdot \|w\|_{k,\alpha} + b_k \cdot \|w\|_{k-1,\alpha}^2,$$

for any integer $k \geq 3$ and some fixed constants $b_2, b_3, \dots$

P Perturbation

1.38. Perturbation lemma. *Let $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$ be a smooth free map. Suppose $L(w) := 2 \cdot Q(f, w)$. Then there is $r > 0$ such that the equation*

$$\textcircled{1} \quad L(w) + Q(w, w) = h$$

has a smooth solution $w: \mathbb{T}^n \rightarrow \mathbb{R}^q$ for any smooth metric h with $\|h\|_{2,\alpha} < r$. Moreover, we can assume that

$$\|w\|_{2,\alpha} \leq C \cdot \|h\|_{2,\alpha},$$

for some constant C .

Suppose that a metric tensor g is induced by a free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$. Note that the equation

$$Q(f + w, f + w) = g + h$$

is equivalent to $\textcircled{1}$. Indeed,

$$\begin{aligned} Q(f + w, f + w) &= Q(f, f) + 2 \cdot Q(f, w) + Q(w, w) \\ &= g + L(w) + Q(w, w). \end{aligned}$$

Therefore the perturbation lemma implies the following.

1.39. Corollary. *Let g_0 be induced by a smooth free map $f_0: \mathbb{T}^n \rightarrow \mathbb{R}^q$ and h another metric tensor on $\mathbb{T}^n$. If $\|h\|_{2,\alpha}$ is sufficiently small, then $g_1 = g_0 + h$ is induced by a smooth map $f_1: \mathbb{T}^n \rightarrow \mathbb{R}^q$.*

Proof of 1.38. Recall that M , a , $\tilde{Q}$, and b_2 are provided by 1.36 and 1.37. Set $R = \frac{1}{10 \cdot b_2}$ and $r = \frac{R}{10 \cdot a}$. Assume $\|h\|_{2,\alpha} \leq r$. By 1.37,

$$\Phi: w \mapsto Mh - \tilde{Q}(w, w)$$

is a contraction map from the closed ball $\bar{B}[0, R] \subset C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ to itself.

Consider the sequence of maps $w_0, w_1, \dots: \mathbb{T}^n \rightarrow \mathbb{R}^q$ such that $w_0 = 0$ and

$$w_{n+1} = \Phi(w_n).$$

Note that w_n is smooth for any n . Since Φ is a contraction, w_n converges in $C^{2,\alpha}(\mathbb{T}^n, \mathbb{R}^q)$ as $n \rightarrow \infty$, say $w_n \rightarrow w$. Further,

$$\|w\|_{2,\alpha} \leq R \quad \text{and} \quad w = Mh - \tilde{Q}(w, w).$$

Since $LM = \text{id}$ and $L\tilde{Q} = Q$, applying L to both sides of the last equation, we get that w solves **1**.

It remains to show that w is smooth. Suppose that

2 *for an integer k , the sequence $\|w_n\|_{k,\alpha}$ is bounded.*

Since

$$\|w\|_{k,\alpha} \leq \lim_{n \rightarrow \infty} \|w_n\|_{k,\alpha}$$

it follows that $w \in C^{k,\alpha}$. Therefore it remains to prove **2** for each integer $k \geq 2$. The base case $k = 2$ is already done; let us do the induction step.

Suppose $\|w_n\|_{k-1,\alpha}$ is bounded; in particular, there is a constant c_k such that

$$\|Mh\|_{k-1,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq c_k$$

for any n ; here b_k is from 1.37.

By the second estimate in 1.37, we get

$$\begin{aligned} \|w_{n+1}\|_{k,\alpha} &\leq \|Mh\|_{k,\alpha} + b_2 \cdot \|w_n\|_{2,\alpha} \cdot \|w_n\|_{k,\alpha} + b_k \cdot \|w_n\|_{k-1,\alpha}^2 \leq \\ &\leq c_k + b_2 \cdot R \cdot \|w_n\|_{k,\alpha} \leq \\ &\leq c_k + \frac{1}{10} \cdot \|w_n\|_{k,\alpha}. \end{aligned}$$

In particular,

$$\|w_n\|_{k,\alpha} \leq 2 \cdot c_k \quad \implies \quad \|w_{n+1}\|_{k,\alpha} \leq 2 \cdot c_k$$

for any n . Since $w_0 = 0$, induction on n implies **2** for k . □

Q Proof of Nash's lemma

Proof of 1.36. Recall that we have global periodic coordinates on $\mathbb{T}^n$.

Choose a metric tensor h on Ω . Since all $\partial_i f$, $\partial_{ij} f$ are linearly independent, the following identities define uniquely a vector field w on $\mathbb{T}^2 \Omega$ that is spanned by $\partial_i f$, $\partial_{ij} f$

$$\langle \partial_i f, w \rangle = 0, \quad \langle \partial_{ij} f, w \rangle = h_{ij}$$

for all i and j .

Since

$$\begin{aligned} L_{ij}(w) &= 2 \cdot Q_{ij}(f, w) = \\ &= \langle \partial_i f, \partial_j w \rangle + \langle \partial_j f, \partial_i w \rangle = \\ &= \partial_j \langle \partial_i f, w \rangle + \partial_i \langle \partial_j f, w \rangle - 2 \cdot \langle \partial_{ij} f, w \rangle, \end{aligned}$$

it defines the needed right inverse $M: h \mapsto w$. □

Let $f: \Omega \rightarrow \mathbb{R}^q$ be a regular smooth map and $p \in \Omega$. Denote by T_p the n -dimensional space spanned by the first-order partial derivatives $\partial_1 f, \dots, \partial_n f$ at p . Denote by $T_p^\perp$ the normal complement of T_p in $\mathbb{R}^q$. We will denote by $T\Omega$ and $T_p^\perp \Omega$ the corresponding bundles over Ω .

Further, denote by T_p^2 the space spanned by all first and second-order partial derivatives $\partial_i f$, $\partial_{ij} f$ at p . Let $N_p = T_p^2 \cap T_p^\perp$.

Note that that if f is free, then T_p^2 and N_p are respectively $(n + \frac{n \cdot (n+1)}{2})$ and $\frac{n \cdot (n+1)}{2}$ -dimensional at any point p . In this case we will denote by $T^2 \Omega$ and $N \Omega$ the corresponding bundles over Ω .

Suppose $w = Mh$ as in the proof above. Since $\langle \partial_i f, w \rangle = 0$ for every i , we have $w(p) \in N_p$ at any point $p \in \Omega$.

1.40. Exercise. Let $f: \Omega \rightarrow \mathbb{R}^q$ be a free map. Given a vector field v in $T\Omega$ and a metric tensor h on Ω , show that there is a unique vector field w in $N\Omega$ such that

$$2 \cdot Q(f, v + w) = h.$$

In other words, given v , there is a right inverse $M_v: h \mapsto v + w$ of L with w in $N\Omega$.

R Proof of Günther's lemma

Note that many symmetric bilinear operators $\tilde{Q}$ satisfy $L\tilde{Q} = Q$. In particular, one can take $\tilde{Q} = MQ$, where M is provided by Nash's

lemma (1.36). However, this choice of $\tilde{Q}$ does not satisfy the inequalities in the Günther lemma. The actual construction of $\tilde{Q}$ uses the extra freedom in choosing M that comes from 1.40.

Recall that we use global system of periodic coordinates on $\mathbb{T}^n$ to define the partial derivatives $\partial_1, \dots, \partial_n$. In particular, we can define the Laplacian

$$\Delta u = \sum_s \partial_{ss} u.$$

The next statement easily follows from the so-called Shauder estimates — a result in elliptic partial differential equations. It could be also obtained directly using Fourier analysis on torus; we omit its proof.

1.41. Proposition. *Let $k \geq 2$ be an integer and $0 < \alpha < 1$. Then*

$$(\Delta - 1): C^{k,\alpha}(\mathbb{T}^n) \rightarrow C^{k-2,\alpha}(\mathbb{T}^n).$$

is a bi-Lipschitz isomorphism. In particular, the linear operator

$$(\Delta - 1): u \mapsto \Delta u - u$$

has a Lipschitz inverse

$$\textcircled{1} \quad (\Delta - 1)^{-1}: C^{k-2,\alpha}(\mathbb{T}^n) \rightarrow C^{k,\alpha}(\mathbb{T}^n)$$

By $\textcircled{1}$, $(\Delta - 1)^{-1}$ is a smoothing operator it swallows $C^{k-2,\alpha}$ and spits $C^{k,\alpha}$. Note that $(\Delta - 1)^{-1}$ commutes with partial derivatives; that is,

$$(\Delta - 1)^{-1} \partial_i = \partial_i (\Delta - 1)^{-1}.$$

These two properties of $(\Delta - 1)^{-1}$ will be important in the following proof.

Proof of 1.37. Recall that $Q_{ij}(w, w) = \langle \partial_i w, \partial_j w \rangle$. Let us show that

$$\textcircled{2} \quad (1 - \Delta)Q_{ij}(w, w) = A_{ij}(w) + \partial_i A_j(w) + \partial_j A_i(w),$$

where each A_{ij} , A_i and A_j is a linear combination of the following quadratic forms

$$\textcircled{3} \quad w \mapsto \langle \partial_a w, \partial_b w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_c w \rangle, \quad w \mapsto \langle \partial_{ab} w, \partial_{cd} w \rangle.$$

for all possible indexes a, b, c, d .

Indeed,

$$\begin{aligned}
 (1 - \Delta)Q_{ij}(w, w) &= \langle \partial_i w, \partial_j w \rangle - \\
 &\quad - \sum_s [2 \cdot \langle \partial_{is} w, \partial_{js} w \rangle + \langle \partial_{iss} w, \partial_j w \rangle + \langle \partial_i w, \partial_{jss} w \rangle] = \\
 &= \langle \partial_i w, \partial_j w \rangle - 2 \cdot \sum_s \langle \partial_{is} w, \partial_{js} w \rangle + 2 \cdot \langle \Delta w, \partial_{ij} w \rangle - \\
 &\quad - \partial_i \langle \Delta w, \partial_j w \rangle - \partial_j \langle \Delta w, \partial_i w \rangle.
 \end{aligned}$$

Whence **2** follows.

Choose $A = A_i$ or $A = A_{ij}$ for some i and j . Note that

$$\begin{aligned}
 \|A(w)\|_{0,\alpha} &\leq c_2 \cdot \|w\|_{2,\alpha}^2, \\
 \|A(w)\|_{k-2,\alpha} &\leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha}^2 + c_k \cdot \|w\|_{k-1,\alpha}^2.
 \end{aligned}
 \tag{4}$$

for some fixed constants $c_2, c_3, \dots$. Indeed, it is sufficient to check inequalities **4** for each form in **3**, and the latter is straightforward.

Applying **1**, we get

$$\begin{aligned}
 \|(1 - \Delta)^{-1}A(w)\|_{2,\alpha} &\leq c_2 \cdot \|w\|_{2,\alpha}^2, \\
 \|(1 - \Delta)^{-1}A(w)\|_{k,\alpha} &\leq c_2 \cdot \|w\|_{k,\alpha} \cdot \|w\|_{2,\alpha} + c_k \cdot \|w\|_{k-1,\alpha}^2;
 \end{aligned}
 \tag{5}$$

the constants $c_2, c_3, \dots$ might be different from those above, but each of them depends only on f .

Further, recall that

$$L_{ij}(w) = -2 \cdot \langle \partial_{ij} f, w \rangle + \partial_i \langle \partial_j f, w \rangle + \partial_j \langle \partial_i f, w \rangle;
 \tag{6}$$

see Section 1Q. Let us define $\tilde{Q}$ as a linear combination of $\partial_i f$ and $\partial_{ij} f$ for all $i \leq j$ such that

$$\begin{aligned}
 -2 \cdot \langle \partial_{ij} f, \tilde{Q}(w, w) \rangle &= (\Delta - 1)^{-1} A_{ij}(w), \\
 \langle \partial_i f, \tilde{Q}(w, w) \rangle &= (\Delta - 1)^{-1} A_i(w)
 \end{aligned}
 \tag{7}$$

for all i and j . Since f is free, $\tilde{Q}$ is well defined. Combining **6**, **7**, and **2**, we get

$$L\tilde{Q} = Q.$$

Denote by e_i and e_{ij} the frame dual to $\partial_i f$ and $\partial_{ij} f$ for all $i \leq j$ in the space spanned by these partial derivatives. Note that

$$\tilde{Q}(w, w) = \sum_i (\Delta - 1)^{-1} A_i(w) \cdot e_i - \frac{1}{2} \cdot \sum_{i \leq j} A_{ij}(w) \cdot e_{ij}.$$

Since the maps $e_i, e_{ij}: \mathbb{T}^n \rightarrow \mathbb{R}^q$ are smooth, **5** and 1.35 imply the last inequality in the lemma for each integer $k \geq 3$. $\square$

S Nash's twist

In this section we prove the following statement which can be regarded as an approximate version of our main theorem (1.28).

1.42. Proposition. *Let g be a Riemannian metric on $\mathbb{T}^n$. Then there is a metric tensor h on $\mathbb{T}^n$ and one-parameter family of smooth maps $w_t: \mathbb{T}^n \rightarrow \mathbb{R}^q$ for $t > 0$ such that*

$$Q(w_t, w_t) = g + t \cdot h$$

for any t .

It might be tempting to pass to the limit of w_t as $t \rightarrow 0$, but, as you will see, this limit is constant; so it does not solve 1.28. The main player in the proof of the proposition is the so-called Nash's twist which are about to define.

Let φ and f be smooth functions on a smooth manifold Ω . Given $r > 0$, denote by $\mathbb{S}_r^1$ the circle of radius r in $\mathbb{R}^2$. The Nash's twist $F: \Omega \rightarrow \mathbb{R}^2$ of the triple (r, φ, f) is the following composition

$$\Omega \xrightarrow{f} \mathbb{R} \xrightarrow{\ell_r} \mathbb{S}_r^1 \xrightarrow{\times \varphi} \mathbb{R}^2,$$

where $\ell_r: \mathbb{R} \rightarrow \mathbb{S}_r^1$ is a length-preserving covering map, say

$$\ell_r(x) = (r \cdot \cos \frac{x}{r}, r \cdot \sin \frac{x}{r}),$$

and $\times \varphi$ denotes the multiplication by φ ; so

$$F(x) := \varphi(x) \cdot (\ell_r \circ f(x)).$$

Recall that $(df)^2$ denotes the metric tensor induced by defined by $f: \Omega \rightarrow \mathbb{R}$; equivalently,

$$(df)^2(x, Y) := df(x) \cdot df(Y) = (xf) \cdot (Yf).$$

1.43. Claim. *The metric tensor*

$$\varphi^2 \cdot (df)^2 + r^2 \cdot (d\varphi)^2$$

is induced by Nash's twist F for (r, φ, f) .

Computations. Suppose that x is a tangent vector on Ω , then

$$\begin{aligned} xF &= x(\varphi \cdot \ell_r \circ f) = \\ &= (x\varphi) \cdot (\ell_r \circ f) + \varphi \cdot (\ell'_r \circ f) \cdot (xf). \end{aligned}$$

Observe that $|\ell_r| = r$, $|\ell'_r| = 1$, and $\ell_r \perp \ell'_r$. Therefore,

$$\langle \mathbf{x}F, \mathbf{y}F \rangle = r^2 \cdot (\mathbf{x}\varphi) \cdot (\mathbf{y}\varphi) + \varphi^2 \cdot (\mathbf{x}f) \cdot (\mathbf{y}f). \quad \square$$

Let $w: \Omega \rightarrow \mathbb{R}^q$ be a smooth map defined on a smooth manifold Ω . Suppose that $w_1, \dots, w_q$ are coordinate functions of w , then the induced metric tensor g can be written as

$$g = Q(v, w) = (dw_1)^2 + \dots + (dw_q)^2.$$

In order to prove 1.28, we need to find functions $w_1, \dots, w_q: \Omega \rightarrow \mathbb{R}$ that solves the above equation for a given Riemannian metric. The following proposition is a weaker form of 1.28; it will play a key role in the proof of 1.42.

1.44. Proposition. *Let (Ω, g) be a compact n -dimensional Riemannian manifold. Then there are smooth functions $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ on Ω such that*

$$\textcircled{1} \quad g = (\varphi_1)^2 \cdot (df_1)^2 + \dots + (\varphi_q)^2 \cdot (df_q)^2.$$

Proof. The metric tensor g can be written in local cart $(x_1, \dots, x_n)$ as

$$\textcircled{2} \quad g = \sum_{i,j} g_{ij} \cdot dx_i \cdot dx_j,$$

where $g_{ij} = g_{ji}$ are smooth functions of $(x_1, \dots, x_n)$.

Since g is Riemannian, we can choose chart so that ∂_i are orthonormal at a given point p ; in other words, $g_{ii} = 1$ and $g_{ij} = 0$ at p for all $i \neq j$. Since g_{ij} are smooth, given $\varepsilon > 0$, we can choose a neighborhood $U \ni p$ such that

$$\textcircled{3} \quad g_{ii} \leq 1 \pm \varepsilon \quad \text{and} \quad g_{ij} \leq \pm \varepsilon$$

for all $i \neq j$ and any point in U .

Observe that

$$\pm dx_i \cdot dx_j = \frac{1}{2} \cdot (dx_i \pm dx_j)^2 - \frac{1}{2} \cdot (dx_i)^2 - \frac{1}{2} \cdot (dx_j)^2.$$

Let us plugin this formula in $\textcircled{2}$ and take $\textcircled{3}$ into account, assuming that ε is small. We get that g is a linear combination with positive coefficients of the metric tensors $(dx_i)^2$ and $(dx_i \pm dx_j)^2$ for all i and j . In other words, we can take the functions x_i and $x_i \pm x_j$ for all i and j as our functions $f_1, \dots, f_q$ and find $\varphi_1, \dots, \varphi_q$ such that $\textcircled{1}$ holds in a neighborhood of p .

By the partition-of-unity, we get the global statement. $\square$

In the presented proof, it is sufficient to take $2 \cdot n^2$ functions in each chart; so, if Ω is covered by m charts, then one can take $q = 2 \cdot m \cdot n^2$ functions. However, as it is stated in the following exercise, one may take q independent of m .

1.45. Exercise. *Show that q in 1.44 can be bounded explicitly in terms of n ; say, we can assume that $q = 100 \cdot n^{100}$.*

Try to generalize 1.44 to noncompact manifolds.

Proof of 1.42. Suppose that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by 1.44. Let $r = \sqrt{t}$; denote by F_i the Nash's twist for the triple (r, φ_i, f_i) . By 1.43,

$$w_t = F_1 \oplus \dots \oplus F_q$$

is the needed map with $h = (d\varphi_1)^2 + \dots + (d\varphi_q)^2$. $\square$

1.46. Advanced exercise. *Show that 1.42 holds for noncompact connected manifolds. Moreover, given any positive smooth function f on Ω , we can assume that h is bounded by $f \cdot g$; that is, $0 \leq h(x, x) \leq f \cdot g(x, x)$ for any tangent vector x on Ω .*

T Proof assembling

1.47. Exercise. *Let Ω be a submanifold in the torus $\mathbb{T}^n$. Show that for any Riemannian metric on Ω is a restriction of a Riemannian metric on $\mathbb{T}^n$.*

Proof of 1.28. By the Whitney embedding theorem, any smooth n -dimensional manifold Ω admits an immersion ι into $\mathbb{R}^{2 \cdot n+1}$. Since Ω is compact, we can assume in addition that ι is short and its image lies in a small ball; if not, then we can pass to the rescaling of $\varepsilon \cdot \iota$ for small $\varepsilon > 0$. Composing ι with the covering map $\mathbb{R}^{2 \cdot n+1} \rightarrow \mathbb{T}^{2 \cdot n+1}$, we get an embedding $\Omega \hookrightarrow \mathbb{T}^{2 \cdot n+1}$.

So from now on, we can assume that Ω is a compact submanifold of a torus.

By 1.47, any Riemannian metric g on Ω is the restriction of a Riemannian metric on the torus. Therefore,

1 *it is sufficient to prove the theorem for $\Omega = \mathbb{T}^n$.*

Choose a free map $f: \mathbb{T}^n \rightarrow \mathbb{R}^q$; it exists by 1.33. Suppose g_1 is the metric tensor induced by f . We can assume that $g_1 < g$; that

is, $g_1(x, x) < g(x, x)$ for any tangent vector $x \neq 0$. In particular, $g_2 = g - g_1$ is a Riemannian metric.

Applying 1.50, we get a metric tensor h on $\mathbb{T}^n$ such that for any $t > 0$ there is a smooth map $f_2: \mathbb{T}^n \rightarrow \mathbb{R}^{q_2}$ that induces $g_2 + t \cdot h$.

By the perturbation lemma, for any small $t > 0$ there is a smooth map $w: \mathbb{T}^n \rightarrow \mathbb{R}^{q_1}$ such that

$$Q(f_1 + w, f_1 + w) = g_1 - t \cdot h.$$

Finally by 1.26, the metric tensor

$$g = g_1 - t \cdot h + g_2 + t \cdot h$$

is induced by the map $(f_1 + w) \oplus f_2$. □

1.48. Exercise. *Let Ω be a connected noncompact manifold. Show that it admits a smooth proper bounded embedding in $\mathbb{R}^q \setminus \{0\}$. Moreover, given a Riemannian metric g on Ω , we can assume that the induced metric of this embedding is strictly smaller than g .*

U Remarks

Let us state another closely related result that shows a huge difference between C^1 and C^2 isometric embeddings. For example, it implies that the unit sphere admits C^1 length-preserving embedding into an arbitrarily small ball in Euclidean 3-space. There is no such C^2 -embedding since Gauss curvature of the unit sphere is 1, but at an extremal point it must be at least $\frac{1}{r^2}$, where r is the radius of the ball.

1.49. Nash–Kuiper theorem. *Let (Ω, g) be a n -dimensional Riemannian manifold and $f: (\Omega, g) \rightarrow \mathbb{R}^q$ be a short smooth regular map. Suppose that $q \geq n + 1$. Then for any $\varepsilon > 0$ there is an C^1 -smooth length-preserving maps $f_\varepsilon: (\Omega, g) \rightarrow \mathbb{R}^q$ that is ε close to f ; that is, $|f_\varepsilon(x) - f(x)| < \varepsilon$ for any $x \in \Omega$.*

Moreover if f is an embedding then we can assume that so is f_ε .

It was originally proved by John Nash [16] with the condition $q \geq n + 2$ and improved to $q \geq n + 1$ by Nicolaas Kuiper [14]. The original proof uses Nash's twist in a different way. Both papers are reader-friendly, it is better to start with the paper of Nash. One may also start with lectures by Allan Yashinski and the author [19] where related results were obtained using an alternative approach.

The discussed result formed a part of foundations of the so-called homotopy principle, or h-principle; an excellent introduction is given in the book by Yakov Eliashberg and Nikolai Mishachev [9].

Many related questions are open. For example, it is unknown if a neighborhood of any point in 2-dimensional Riemannian manifold admits a smooth length-preserving embedding into $\mathbb{R}^3$.

This theorem is a leading partial case of Nash theorem on regular emebdding. The compactness condition can be removed; we will give a hint how to do it. Moreover the value q can be estimated in terms $n = \dim \Omega$. We will give a hint how to get a ruf bound, say $q = 100 \cdot n^3$. The best known estimate so far is

$$q = \max\left\{\frac{n \cdot (n+5)}{2}, \frac{n \cdot (n+3)}{2} + 5\right\}.$$

In the following exercises we assume that $\varphi_1, \dots, \varphi_q, f_1, \dots, f_q$ are provided by 1.44 for a given Riemannian manifold (Ω, g) . Further, F_i is the Nash's twist for the triple (r, φ_i, f_i) , and

$$\begin{aligned} h &= (d\varphi_1)^2 + \dots + (d\varphi_q)^2, \\ F_r &= F_1 \oplus \dots \oplus F_q: \Omega \rightarrow \mathbb{R}^{2 \cdot q}, \\ \Phi &= \varphi_1 \oplus \dots \oplus \varphi_q: \Omega \rightarrow \mathbb{R}^q. \end{aligned}$$

1.50. Exercise. Let (Ω, g) be a compact n -dimensional Riemannian manifold and $r > 0$.

- (a) Show that $g + r^2 \cdot h$ is induced by F_r .
- (b) Show that $F_r(p) \rightarrow 0$ as $r \rightarrow 0$ for any point $p \in \Omega$.

Let us denote by $\mathbb{R}^{r,s}$ the pseudoeuclidean space with signature (r, s) ; that is, the space $\mathbb{R}^{r+s}$ with scalar product defined by

$$\langle x, y \rangle = x_1 \cdot y_1 + \dots + x_s \cdot y_s - x_{s+1} \cdot y_{s+1} - \dots - x_{s+r} \cdot y_{s+r},$$

where x_i and y_i denote the coordinates of vectors x and y in $\mathbb{R}^{r+s}$.

Show that g is induced by

$$F_r \oplus \Phi: \Omega \rightarrow \mathbb{R}^{2 \cdot q, q} = \mathbb{R}^{2 \cdot q, 0} \oplus \mathbb{R}^{0, q}.$$

The presented construction can be used to find an isometric immersion of Riemannian manifold (Ω, g) into a pseudoeuclidean space $\mathbb{R}^{r+s}$ with sufficiently large r and s .

1.51. Exercise. Let $f: (\Omega, g) \rightarrow \mathbb{S}^{q-1}$ be a smooth length-preserving embedding. Construct a smooth length-preserving embedding of any conformally equivalent manifold $(\Omega, \varphi^2 \cdot g)$ into $\mathbb{R}^{q,1}$.

That is, given a smooth positive function φ on Ω , construct a smooth map $\Omega \rightarrow \mathbb{R}^{q,1}$ with induced metric tensor $\varphi^2 \cdot g$.

Lecture 2

Algebra of curvature

The curvature of a Riemannian manifold is described by a tensor, not just a number. This is one of the principle differences between differential geometry of surfaces and higher-dimensional differential geometry.

In this lecture we will give an outline of algebra related to curvature tensor. Most of the statements come without proofs, but everything can be proved by straightforward calculations (which are often tedious).

Most proofs can be found in [8, Chapters 4+6] and [4, Chapter 3].

A Definition

Let x , y , v , and w be vector fields on a Riemannian manifold (M, g) . Recall that ∇ denotes the Levi-Civita connection on M .

The Riemannian curvature tensor R is defined by¹

$$R(x, y)v = \nabla_x \nabla_y v - \nabla_y \nabla_x v - \nabla_{[x, y]} v.$$

It has valence 4 — it takes 3 vectors and spits another vector. We do not need to specify covariance/contravariance type of the tensor since the metric tensor identifies tangent and cotangent bundles.

By the definition, one sees that the curvature tensor depends linearly on vector fields. But it is indeed a tensor — that is, the vector $R(x, y)v$ depends only on the tangent vectors x , y , and v at the point.

¹Many authors (for example do Carmo) define it with opposite sign. If you see notation Rm then most likely the sign is opposite. This convention fits better with an earlier convention that sphere has positive Gauss curvature.

The latter follows from the following identities:

$$\begin{aligned} f \cdot R(X, Y)(V) &= R(f \cdot X, Y)(V) = \\ &= R(X, f \cdot Y)(V) = \\ &= R(X, Y)(f \cdot V). \end{aligned}$$

for any vector fields X, Y, V and a function f . All of them can be proved by straightforward computations.

B Curvature transformation

For given tangent vectors X and Y at a point the linear map

$$R(X, Y): T \rightarrow T$$

is called curvature transformation; some authors prefer to denote it by $R_{X,Y}$. It has the following geometric meaning:

Let γ be the contour of small parallelogram spanned by vectors X and Y at a point p . Let us denote by $\iota_\gamma: T_p \rightarrow T_p$ the parallel transport along γ . Denote by a the area of parallelogram. Then

$$\iota_\gamma = \text{id} + a \cdot R(X, Y) + o(a).$$

where id denotes the identity map on T_p .

2.1. Exercise. *Suppose that parallel transport $T_p \rightarrow T_q$ in a Riemannian manifold (M, g) does not depend on a path connecting p to q . Show that (M, g) is flat; that is, its curvature tensor vanishes at all points.*

C Symmetries

Set

$$\hat{R}(X, Y, V, W) := \langle R(X, Y)V, W \rangle.$$

Note that $\hat{R}$ remembers everything about the curvature tensor R (assuming that metric tensor is known). The $\hat{R}$ -form of R is more convenient to describe the symmetries of curvature tensor:

$$\begin{aligned} \textcircled{1} \quad \hat{R}(X, Y, V, W) &= -\hat{R}(Y, X, V, W) = -\hat{R}(X, Y, W, V), \\ 0 &= \hat{R}(X, Y, V, W) + \hat{R}(Y, V, X, W) + \hat{R}(V, X, Y, W). \end{aligned}$$

The last identity is called the algebraic Bianchi identity or first Bianchi identity (and it was not discovered by Bianchi).

These identities can be proved by straightforward computations. Latter we will show that these symmetries provide a complete list; that is, given a tensor that satisfies these identities, there is a Riemannian manifold with such curvature tensor at some point.

The following equality follows from the main symmetries

$$\textcircled{2} \quad \hat{R}(x, y, v, w) = \hat{R}(v, w, x, y).$$

D Space of curvature tensors

Let us denote by $A^4 T$ the space of all algebraic curvature tensors on T ; that is, $A^4 T$ all valence 4 tensors with the symmetries $\textcircled{1}$.

Given a Euclidean space E , we denote by $S^n E$ and $\Lambda^n E$ the space of symmetric and antisymmetric tensors of valence n over E .

Note that $\textcircled{2}$ and the first line in $\textcircled{1}$ imply that

$$A^4 T \subset S^2 \Lambda^2 T.$$

In other words, a curvature tensor can be described as a symmetric bilinear form on the space of bivectors $\Lambda^2 T$; or as the so-called curvature operator — a linear operator $R: \Lambda^2 T \rightarrow \Lambda^2 T$ defined by²

$$\langle R(x \wedge y), v \wedge w \rangle = -\langle R(x, y)v, w \rangle.$$

The symmetry $\textcircled{2}$ implies that R is self-adjoint; that is,

$$\langle R\varphi, \psi \rangle = \langle \varphi, R\psi \rangle$$

for any bivectors $\varphi, \psi \in \Lambda^2 T$.

The algebraic Bianchi identity implies that complete antisymmetrization of $\hat{R}$ vanishes. More precisely, if $\alpha: S^2 \Lambda^2 T \rightarrow \Lambda^4 T$ denotes the complete antisymmetrization then the space of curvature tensors is the kernel of α .³ The latter can be written as

$$A^4 T = S^2 \Lambda^2 T \ominus \Lambda^4 T \quad \text{or} \quad A^4 T = S^2 \Lambda^2 T \cap (\Lambda^4 T)^\perp,$$

²If $\{e_i\}$ is an orthonormal basis of T , then the scalar product on $\Lambda^2 T$ is defined by stating that it has an orthonormal basis $\{e_i \wedge e_j\}_{i > j}$.

Alternatively, the scalar product in $\Lambda^2 T$ can be also defined on simple bivectors $x \wedge y$ and $v \wedge w$ by stating that

$$\langle x \wedge y, v \wedge w \rangle = \langle x, v \rangle \cdot \langle y, w \rangle - \langle x, w \rangle \cdot \langle y, v \rangle$$

and extended linearly to whole $\Lambda^2 T$.

³As far as I see, the following property is absolutely useless, but it is funny. Given a curvature tensor R consider the tensor

$$\mathfrak{A}(x, v, y, w) := \langle R(x, y)v, w \rangle + \langle R(v, y)x, w \rangle.$$

Then the linear transformation $R \rightarrow \mathfrak{A}$ describes an isomorphism

$$S^2 \Lambda^2 T \ominus \Lambda^4 T \longleftrightarrow S^2 S^2 T \ominus S^4 T.$$

where $L^\perp$ stands for the orthogonal complement L in the Euclidean metric on $S^2\Lambda^2 T$ induced from T .

If $n = \dim T$ then the dimension of the space of curvature tensors over T can be easily calculated:

$$\dim(S^2\Lambda^2 T) - \dim(\Lambda^4 T) = \binom{\binom{n}{2} + 1}{2} - \binom{n}{4} = \frac{n^2 \cdot (n^2 - 1)}{12}.$$

2.2. Exercise. Suppose $\mathbf{R}_\varphi : \Lambda^2 T \rightarrow \Lambda^2 T$ is an orthogonal projection to a 1-dimensional subspace spanned by a bivector $\varphi \in \Lambda^2 T$. Show that $\mathbf{R}$ is a curvature operator if and only if φ is a simple bivector; that is, if $\varphi = X \wedge Y$ for some $X, Y \in T$.

Show that in this case $\mathbf{R}$ is the curvature operator of $\mathbb{S}^2 \times \mathbb{R}^{n-2}$.

Use the results in Section 2H to show that any algebraic curvature tensor can appear as a curvature tensor of a Riemannian manifold.

E Sectional curvature

Let p be a point in a Riemannian manifold (M, g) . Choose a plane σ in the tangent space T_p . Consider a surface Σ in M swept by short geodesics from p in the directions on σ . The Gauss curvature of Σ at p is called sectional curvature and denoted by $\sec \sigma$; the plane σ is called sectional direction.

If the sectional direction σ is spanned by vectors X and Y , then

$$\begin{aligned} \sec \sigma &= \frac{\langle \mathbf{R}(X, Y)Y, X \rangle}{|X|^2 \cdot |Y|^2 - \langle X, Y \rangle^2} = \\ &= \frac{K(X, Y)}{|X \wedge Y|^2}; \end{aligned}$$

in the last expression we use shortcut

$$K(X, Y) = \langle \mathbf{R}(X, Y)Y, X \rangle = -\langle \mathbf{R}(X, Y)X, Y \rangle = \langle \mathbf{R}(X \wedge Y), X \wedge Y \rangle;$$

note that K is quadratic in both arguments.

The formula above implies that sectional curvature can be recovered from curvature tensor.

That is, curvature tensor can be described as quadratic forms on quadratic forms that lie in the kernel of complete symmetrization $\sigma : S^2 S^2 T \rightarrow S^4 T$; in other words $\mathbf{R}$ satisfies the following symmetries:

$$\begin{aligned} \mathbf{R}(x, y, z, w) &= \mathbf{R}(y, x, z, w) = \mathbf{R}(x, y, w, z) = \mathbf{R}(y, w, x, z), \\ 0 &= \mathbf{R}(x, x, x, x). \end{aligned}$$

The curvature tensor can be expressed using sectional curvature as well. Indeed, once we know curvature in any sectional direction, we can use the above formula to find $K(x, y)$ for any tangent vectors $x, y \in T_p$. After that one could apply the following formula:

$$\begin{aligned} 6 \cdot R(x, y)v, w) = & [K(x + v, y + w) + K(x, w) + K(y, v) - \\ & - K(x + v, y) - K(x + v, w) - K(x, y + w) - K(v, y + w)] - \\ & - [K(x + w, y + v) + K(y, w) + K(x, v) \\ & - K(x + w, y) - K(x + w, v) - K(x, y + v) - K(w, y + v)]. \end{aligned}$$

The formula is scary, but it is very similar to recovery of a symmetric bilinear form its quadratic form.

2.3. Exercise. Let $\dim T = 3$. Suppose that a curvature tensor $R \in A^4 T$ has positive sectional curvature in all directions. Show that the corresponding curvature operator is positive; that is $\langle R\varphi, \varphi \rangle > 0$ is for any $\varphi \in \Lambda^2 T$.

Show that if $\dim T = 4$, then analogous statement does not hold.

F Ricci decomposition

Let E_i be an orthonormal basis at a point p of Riemannian manifold. The so called Ricci curvature tensor is a linear transformation, $\text{Ric}: T_p \rightarrow T_p$ defined as

$$\langle \text{Ric } x, x \rangle = \sum_i K(x, E_i).$$

Further, the scalar curvature Sc is defined by

$$Sc = \sum_{i,j} K(E_i, E_j) = \sum_j \langle \text{Ric } E_j, E_j \rangle = 2 \cdot \text{trace } R.$$

If $|x| = 1$ then the value $\langle \text{Ric}(x), x \rangle$ is called Ricci curvature in the direction x . For example n -dimensional unit sphere has sectional curvature 1, Ricci curvature $n - 1$ in all directions and scalar curvature $n \cdot (n - 1)$.

The action of orthogonal group $O(T)$ can be extended to $A^4 T$. Evidently the kernels of $R \rightarrow \text{Ric}$ and $R \rightarrow Sc$ and their orthogonal complements are invariant with respect to this action. In other words, Ricci tensor Ric and scalar curvature Sc do not depend on the choice of the orthonormal basis E_i .

It turns out that these are the only subspaces of $A^4 T$ that are invariant with respect to $O(T)$. In other words, the decomposition

$$A^4 T = U \oplus V \oplus W$$

with the subspaces U , V , and W described below is the maximal $O(T)$ -invariant decomposition of $A^4 T$.

- ◊ W is the kernel of $R \rightarrow \text{Ric}$; the orthogonal projection to W of the curvature tensor is called its Weil tensor.
- ◊ V is the intersection of the kernel $R \rightarrow \text{Sc}$ and the orthogonal complement of W . The projection to V is completely determined by the traceless Ricci tensor $\text{Ric}_0 = \text{Ric} - \frac{1}{n} \cdot \text{Sc} \cdot g$.
- ◊ U is a one-dimensional subset of curvature tensors with constant sectional curvatures; so the curvature operator is proportional to the identity.

G Curvature bounds

The following theorem is a classical result in Riemannian geometry with long history.

2.4. Quarter-pinched sphere theorem. *Suppose that (M, g) is a simply connected Riemannian manifold with sectional curvature strictly between 1 and 4 at each point. Then M is diffeomorphic to a sphere.*

It gives an example of the so called local-to-global theorems. Its assumption is a local property that is given by a curvature bounds at each point. The conclusion says something about global structure of the manifold; in this example it says something about its topological type.

Riemannian geometry has other types of theorems⁴, but nothing else makes Riemannian geometry nearly as happy as the local-to-global results.

Typically the local condition is given by restriction on curvature tensor of Riemannian manifold; that is, we specify a subset $\Omega \subset A^4 T$ and assume that curvature tensor of a Riemannian manifold belongs to Ω at each point. Most of the time the set $\Omega \subset A^4 T$ is open or at least has nonempty interior.⁵

It is reasonable to assume that Ω is convex (or at least connected). If $\dim T = 2$, then $\dim A^4 T = 1$; in this case we do not have much choice — we might consider curvature bounded above, or below, or from both sides. That what we did for surfaces.

Starting from dimension 3, things getting more complicated — it is reasonable to assume in addition that Ω is invariant with respect to

⁴for example rigidity theorems say that a Riemannian manifold with given property must be isometric to some known manifold (typically a round sphere).

⁵There are exceptions, for example Einstein manifolds defined by an equation on curvature tensor. But this subject lies on a half way from differential geometry to partial differential equations.

rotations of the tangent space; in other words Ω has to be invariant with respect to action of the orthonormal group $O(T)$ of the space T extended to $A^4 T$. But still, we have huge family of curvature conditions.

The Ricci flow technique deals with few families of curvature bounds in the proofs. Couple of dozens of curvature bounds made it to a formulation a meaningful theorem. The champions seem to be upper, lower, and bilateral bounds on sectional curvature, lower bounds on Ricci curvature, and lower bounds on scalar curvature.⁶

H Submanifolds

Suppose M is a submanifold in a Riemannian manifold $(\bar{M}, g)$. Recall that restricting g to the tangent bundle over M produce a Riemannian metric on M . Let us denote by $\bar{\nabla}$ and ∇ the Levi-Civita connection on $\bar{M}$ and M respectively.

Recall that second fundamental form S of M is defined by

$$S(X, Y) = \nabla_X Y - \bar{\nabla}_X Y;$$

it takes two tangent vectors on M and spits a normal vector; so is a tensor in $S^2 TM \otimes NM$ (here we identify tangent/cotangent normal/conormal bundles as usual).

Again straightforward computations show that that S is indeed a tensor — one has to show that

$$f \cdot S(X, Y) = S(f \cdot X, Y) = S(X, f \cdot Y)$$

for any tangent vector fields X, Y , and a smooth function f on M .

The following formula gives a relation between curvature tensors R and $\bar{R}$ of M and $\bar{M}$ respectively:

$$\begin{aligned} \langle R(X, Y)Z, W \rangle &= \langle \bar{R}(X, Y)Z, W \rangle + \\ &+ \langle S(X, W), S(Y, Z) \rangle - \langle S(X, Z), S(Y, W) \rangle. \end{aligned}$$

In particular, using the shortcut $K(X, Y) = \langle R(X, Y)Y, X \rangle$, we can write

$$K(X, Y) = \bar{K}(X, Y) + \langle S(X, X), S(Y, Y) \rangle - |S(X, Y)|^2.$$

This is a generalization of the formula for Gauss curvature of surface in the Euclidean space:

$$K = \ell \cdot n - m^2,$$

⁶As far as I see, the following property is absolutely useless, but it is funny. The cone of curvature tensors with nonnegative (or nonpositive) sectional curvature is a maximal $GL(T)$ -invariant subsets of $A^4(T)$.

where ℓ , m , and n are components of Hessian matrix at p in an orthonormal basis. Indeed, if X, Y is an orthonormal frame at a point in a surface, then $K = K(X, Y)$ is its Gauss curvature, and

$$S(X, X) = \ell \cdot \nu, \quad S(X, Y) = m \cdot \nu, \quad S(Y, Y) = n \cdot \nu,$$

where ν is a unit normal vector at p .

2.5. Exercise. Show that there is a 4-dimensional Riemannian manifold (M, g) such that no neighborhood of a point p in M admits a smooth length-preserving embedding in $\mathbb{R}^5$.

Hint: Count dimensions of second fundamental form and curvature tensor at p and apply the formula.

2.6. Exercise. Let M be a smooth submanifold of a Euclidean space. Assume $\text{codim } M = 2$. Show that if sectional curvature of M is positive, then so is its curvature operator.

An analogous statement for submanifolds of codimension 3 does not hold — try to guess why.

Hint: Show first that $\langle S(X, X), S(Y, Y) \rangle > 0$ for any two nonvanishing tangent vectors X, Y at any point $p \in M$. Further, show and use that if $\text{codim } M = 2$, then the normal space admits an orthonormal basis U_1, U_2 such that both quadratic forms $s_i(X, Y) := \langle S(X, Y), U_i \rangle$ are positive definite.

I Submersions

Let $s: \bar{M} \rightarrow M$ be a submersion between smooth manifolds. Suppose that manifolds $\bar{M}$ and M are equipped with Riemannian metrics.

Suppose that $s(\bar{p}) = p$.

The kernel of the differential $ds: T_{\bar{p}}\bar{M} \rightarrow T_p M$ will be called vertical subspace; it will be denoted by $V_{\bar{p}}$. The orthogonal complement of $V_{\bar{p}}$ in $T_{\bar{p}}\bar{M}$ will be called horizontal subspace at $\bar{p}$; it will be denoted by $H_{\bar{p}}$.

The submersion s is called Riemannian if the restriction of ds to H is an isometry.

Note that $T_{\bar{p}} = H_{\bar{p}} \oplus V_{\bar{p}}$. In particular, any tangent vector $X \in T_{\bar{p}}\bar{M}$ can be split into its vertical and horizontal part denoted by X^H and X^V , so

$$X = X^H + X^V.$$

By the definition of Riemannian submersion, for any vector $X \in T_p M$ there is a unique vector $\bar{X} \in H_{\bar{p}}$ such that $X = ds(\bar{X})$.

The curvature of M can be found using the so called O'Neil formula:

$$K(X, Y) = \bar{K}(\bar{X}, \bar{Y}) + \frac{3}{4} \cdot |[\bar{X}, \bar{Y}]^V|^2.$$

The notion of Riemannian submersion is dual to submanifold ($\approx$ length-preserving immersion). The tensor A defined by

$$A(\bar{X}, \bar{Y}) = [\bar{X}, \bar{Y}]^V$$

is indeed a tensor in $\Lambda^2 H \otimes V$ (it is proved the usual way). This tensor plays a role similar to the second fundamental form of a submanifold.

2.7. Exercise. Let $\bar{M} \rightarrow M$ be a Riemannian submersion. Suppose that $\bar{M}$ has nonnegative curvature operator at all points. Show that at any point of M there is a 4-vector η such that

$$\langle R\varphi, \varphi \rangle + \langle \eta, \varphi \wedge \varphi \rangle \geq 0$$

for any 2-vector φ .

Hint: Use Section 2D.

J Lie groups

Suppose G is a Lie group with biinvariant metric. By straightforward computations, one gets the following identities for left-invariant vector fields on G :

$$\begin{aligned} \nabla_X Y &= \frac{1}{2} \cdot [X, Y] \\ \langle R(X, Y)Z, W \rangle &= \frac{1}{4} \cdot (\langle [X, W], [Y, Z] \rangle - \langle [X, Z], [Y, W] \rangle) \\ K(X, Y) &= \frac{1}{4} \cdot |[X, Y]|^2 \end{aligned}$$

Note that the first identity implies that $\nabla_X X = 0$. Therefore homomorphisms $\mathbb{R} \rightarrow G$ are geodesics.

K Cheeger's trick

Suppose a Lie group G acts isometrically on a Riemannian manifold $M = (M, g)$. Suppose that G admits a bi-invariant metric (this is always the case if G is compact).

Consider the diagonal action of G on the product $G \times M$; that is, $a \cdot (b, x) := (a \cdot b, a \cdot x)$ for any $a, b \in G$ and $x \in M$. Note that this action is isometric and free. Therefore the quotient map $(\lambda \cdot G) \times M \rightarrow (M, g_\lambda)$ is a Riemannian submersion for some metric g_λ . This way we obtain

a one parameter family g_λ of Riemannian metrics on M . Note that $g_\lambda \rightarrow g$ as $\lambda \rightarrow \infty$.

This procedure is called Cheeger's trick and the obtained family is called Cheeger's deformation. (Jeff Cheeger found number of applications of this trick, but it was invented earlier.)

By O'Neil formula, if (M, g) had a nonnegative (respectively positive) sectional curvature, then so does (M, g_λ) for any $\lambda > 0$.

L Berger spheres

Applying the Cheegers to the isometric action of $\mathbb{S}^1$ on $\mathbb{S}^3$ by complex multiplication one gets Berger spheres — a family of metrics g_λ on $\mathbb{S}^3$ with positive sectional curvature.

The Berger spheres can be also described as certain left-invariant metrics on the Lie group $\mathbb{S}^3 = \text{Spin}(4)$. There are explicit formulas for connection and curvature in such metrics.

It is an important example in Riemannian geometry. (It could be compared to the Cantor set in analysis.)

2.8. Exercise. Suppose $(\mathbb{S}^3, g_\lambda)$ denotes Berger spheres with parameter λ . Set $\ell = \lambda/\sqrt{1+\lambda^2}$.

Show that

(a) $(\mathbb{S}^3, g_\lambda)$ are foliated by closed geodesics of length $2\pi\ell$ and

$$\text{vol}(\mathbb{S}^3, g_\lambda) = \ell \cdot \text{vol } \mathbb{S}^3.$$

(b) Show that the curvature tensor of $(\mathbb{S}^3, g_\lambda)$ converges to the curvature tensor of $\mathbb{S}^2_{\frac{1}{2}} \times \mathbb{R}$ as $\lambda \rightarrow 0$, where $\mathbb{S}^2_{\frac{1}{2}}$ denotes a round sphere of radius $\frac{1}{2}$. (One has to find right sense for term “converge” here.)

(c) Try to visualize what happens with $(\mathbb{S}^3, g_\lambda)$ as $\lambda \rightarrow 0$.

Lecture 3

Second variation

A Exercises

Use the formulas from the next section to solve the following exercises:

3.1. Exercise. Let M be a closed n -dimensional Riemannian manifold with injectivity radius at least π at each point.

- (a) Choose two orthonormal vectors $V, T \in T_p M$. Consider geodesic $\gamma: [0, \pi]$ in the direction of T . Denote by $V(t), T(t) \in T_{\gamma(t)}$ the parallel translations of V and T along γ , so $T = \gamma'$. Denote by $\kappa(V, T)$ the average value of $\sec(V(t) \wedge T(t)) \cdot \sin^2(t)$ for $t \in [0, \pi]$. Show that

$$\kappa(V, T) \leq \frac{1}{2}$$

for any $p \in M$ and any orthonormal pair $V, T \in T_p M$.

- (b) Apply Liouville's theorem to show that the average of sectional curvature on M is at most 1. Conclude that M has a point with scalar curvature at most $n \cdot (n - 1)$.

3.2. Exercise (Syng's theorem). Let γ be a closed simple geodesic on Riemannian manifold M . Suppose that either a neighborhood of γ is orientable and dimension of M is even, or a neighborhood of γ is nonorientable and dimension of M is odd.

- (a) Show that there is a parallel unit vector field W on γ that is orthogonal to γ . Conclude that γ admits a length-decreasing homotopy.
- (b) Use part (a) to show that if M is oriented and has even dimension, then it is simply connected.
- (c) Use part (a) to show that if M has odd dimension, then it is oriented.

A submanifold M of Riemannian manifold $\bar{M}$ is called totally geodesic if for every point $p \in M$ and any tangent vector $v \in T_p M$, the geodesic γ with initial value $\gamma'(0) = v$ lies in M .

3.3. Exercise (Frankel's theorem). *Let M and N be two totally geodesic submanifolds in a closed n -dimensional Riemannian manifold with positive sectional curvature. Suppose that $\dim M + \dim N > n$. Show that M meets N at some point.*

B Equations

Jacobi equation. Suppose J is a Jacobi field along a geodesic γ and $T = \gamma'$. Then

$$J'' + R(J, T)T = 0;$$

here we use shortcut notation $J' = \nabla_T J$.

Riccati equation. Let γ be a unit-speed geodesic. Suppose a smooth function f is defined in its neighborhood and satisfies the following conditions:

$$f \circ \gamma(t) \equiv t, \quad |\nabla f| \equiv 1.$$

Set $T = \nabla f$, note that this vector field extends γ' . Let S be the shape operator of the level sets of f ; that is

$$S(v) = \nabla_v T.$$

Then the following identity holds:

$$S' + S^2 + R(\cdot, T)T = 0;$$

as before we write S in a parallel frame on γ , so S' is a shortcut for $\nabla_T S$. In other words, if v is a parallel vector field along γ , then

$$S'(v) + S^2(v) + R(v, T)T = 0.$$

Second variation formula. Let w be a vector field normal to geodesic $\gamma: [a, b] \rightarrow M$. As before $T = \gamma'$ and w' is a shortcut for $\nabla_T w$.

Family of curves $\gamma_t = \exp_{\gamma(t)}(t \cdot w)$, set $L(t) = \text{length } \gamma_t$. Then

$$L''(0) = \int_a^b |w'|^2 - K(w, T).$$

Bibliography

- [1] A. Besicovitch. “On two problems of Loewner.” *J. Lond. Math. Soc.* 27 (1952), 141–144.
- [2] G. Besson, G. Courtois, and S. Gallot. “Entropies et rigidités des espaces localement symétriques de courbure strictement négative”. *Geom. Funct. Anal.* 5.5 (1995), 731–799.
- [3] D. Burago and S. Ivanov. “Boundary rigidity and filling volume minimality of metrics close to a flat one”. *Ann. of Math. (2)* 171.2 (2010), 1183–1211.
- [4] J. Cheeger and D. Ebin. *Comparison theorems in Riemannian geometry*. Revised reprint of the 1975 original. AMS Chelsea Publishing, Providence, RI, 2008, x+168.
- [5] C. B. Croke. “Rigidity and the distance between boundary points”. *J. Differential Geom.* 33.2 (1991), 445–464.
- [6] C. B. Croke and B. Kleiner. “Conjugacy and rigidity for manifolds with a parallel vector field”. *J. Differential Geom.* 39.3 (1994), 659–680.
- [7] C. B. Croke and B. Kleiner. “A rigidity theorem for simply connected manifolds without conjugate points”. *Ergodic Theory Dynam. Systems* 18.4 (1998), 807–812.
- [8] M. do Carmo. *Riemannian geometry*. Mathematics: Theory & Applications. Translated from the second Portuguese edition by Francis Flaherty. Birkhäuser Boston, Inc., Boston, MA, 1992, xiv+300.
- [9] Y. Eliashberg and N. Mishachev. *Introduction to the h-principle*. Vol. 48. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2002, xviii+206.
- [10] M. Gromov. “Filling Riemannian manifolds”. *J. Differential Geom.* 18.1 (1983), 1–147.
- [11] M. Günther. “On the perturbation problem associated to isometric embeddings of Riemannian manifolds”. *Ann. Global Anal. Geom.* 7.1 (1989), 69–77.
- [12] R. Howard. *Notes on Günther’s Method and the Local Version of the Nash Isometric Embedding Theorem*. URL: <https://people.math.sc.edu/howard/Notes/nash.pdf>.
- [13] С. В. Иванов. *Некоторые геометрические неравенства и теоремы жесткости*. Математическая лаборатория имени П.Л. Чебышева. 2012. URL: <https://www.lektorium.tv/lecture/13892>.
- [14] Nicolaas H. Kuiper. “On C^1 -isometric imbeddings. I, II”. *Nederl. Akad. Wetensch. Proc. Ser. A.* 58 = *Indag. Math.* 17 (1955), 545–556, 683–689.
- [15] R. Michel. “Sur la rigidité imposée par la longueur des géodésiques”. *Invent. Math.* 65.1 (1981/82), 71–83.
- [16] J. Nash. “ C^1 isometric imbeddings”. *Ann. of Math. (2)* 60 (1954), 383–396.
- [17] J. Nash. “The imbedding problem for Riemannian manifolds”. *Ann. of Math. (2)* 63 (1956), 20–63.

- [18] L. Pestov and G. Uhlmann. “Two dimensional compact simple Riemannian manifolds are boundary distance rigid”. *Ann. of Math. (2)* 161.2 (2005), 1093–1110.
- [19] A. Petrunin and A. Yashinski. “Piecewise isometric mappings”. *Algebra i Analiz* 27.1 (2015), 218–247.
- [20] T. Tao. *Notes on the Nash embedding theorem*. URL: <https://terrytao.wordpress.com/2016/05/11/>.
- [21] D. Yang. *Gunther’s proof of Nash’s isometric embedding theorem*. 1998.